TRITON — COMPARATIVE SYSTEMS STUDY

# TECHNICAL AND SCIENTIFIC RESEARCH REPORT

Integrated comparison of rainwater harvesting configurations: hydraulic performance, economic and financial differences, environmental impacts, and multi-criteria trade-offs

| | |
|---|---|
| **Reference code:** | N5-20261015-085015-27731E68 |
| **Project name:** | Example - tutorial house in New York (ERA5 1991-2020) [costs to be entered by the user] |
| **Standards basis:** | ISO 14040/14044 |
| **Issue date:** | 15/10/2026 |
| **Program edition:** | International Edition (INT) |
| **Technical responsibility:** | Responsible Researcher / Engineer |

2026

## LIST OF GENERATED FIGURES

The figures below were calculated by Triton from the same inputs and alternatives presented in the tables. Numbering follows the chapter reading order.

| Nº | Figure | Thematic chapter |
|---|---|---|
| **1** | Hydraulic performance: water-saving potential | Hydraulic performance |
| **2** | Hydraulic performance: marginal gain | Hydraulic performance |
| **3** | Hydraulic performance: rainwater use and overflow | Hydraulic performance |
| **4** | Hydraulic performance: annual supply reliability | Hydraulic performance |
| **5** | Multi-criteria decision: TOPSIS performance | Multi-criteria decision |
| **6** | Multi-criteria decision: proximity by dimension | Multi-criteria decision |
| **7** | Economic assessment: NPV sensitivity | Economic assessment |
| **8** | Economic assessment: NPV by capacity | Economic assessment |

# TECHNICAL AND SCIENTIFIC RESEARCH REPORT

Integrated assessment of water performance, economic feasibility and life-cycle assessment (LCA).

| REFERENCE CODE | ISSUE DATE | STANDARDS BASIS | MOTOR / ENGINE |
|---|---|---|---|
| N5-20261015-085015-27731E68 | 15/10/2026 | ISO 14040/14044 | Triton (YAS) |
| **LOCATION** | **CLIMATE BASIS** | **SERIES PERIOD** | **EMBEDDED FIGURES** |
| New York / USA | INM-CM4 · HIST | 01/01/1991 a 31/12/2020 | 8 |

| REPRESENTATIVE SYSTEMS | TANK RANGE | NPV RANGE | CLIMATE BALANCE RANGE |
|---|---|---|---|
| **3** | **1,000-14,000 L** | **$ 919 a 1,343** | **-626.3 a -163.7 kg** |
| distinct configurations | objective-based references | reported horizon | CO2e over the life cycle |

📌 **Guidelines on Scientific Report Structure**

The report starts with differences among systems and then examines the hydraulic, economic and environmental dimensions. Technical, economic and environmental references are criterion-specific results; the TOPSIS compromise is the alternative closest to the weighted ideal, not an absolute optimum.

The report includes **8 calculated figures**, distributed across thematic chapters with captions, interpretation, and source.

# 1. COMPARATIVE DECISION SUMMARY

📌 **Executive Water Engineering & Multi-Criteria Synthesis**

This report consolidates the technical-scientific analysis of the rainwater harvesting system, integrating continuous hydrological simulation (YAS Model — Yield After Spillage), economic-financial feasibility via discounted cash flow, and simplified Life Cycle Assessment (LCA). The data provides complete traceability of project assumptions according to the technical standards and regulations applicable at the project site. Decision-making balances technical supply reliability, financial returns, and greenhouse gas (GHG) emission mitigation. Rather than declaring one universally optimal solution, the report compares storage configurations and makes the consequences of each decision criterion explicit.

## 1.1 Comparison of decision references

| Decision role | Tank | Savings | Rainwater use | CAPEX | VPL / NPV | CO2e | TOPSIS |
|---|---|---|---|---|---|---|---|
| **Technical reference + TOPSIS compromise**<br>highest observed water-saving potential; highest weighted closeness to the ideal | **4,000 L** | 26.8% | 58.8 $m^3$/year | $ 0.00 | $ 1,288.12 | -271.4 kg CO2e | 0.912 |
| **Economic reference**<br>highest NPV over the horizon | **14,000 L** | 28.0% | 61.3 $m^3$/year | $ 0.00 | $ 1,343.38 | -626.3 kg CO2e | 0.840 |
| **Environmental reference**<br>best life-cycle climate balance | **1,000 L** | 19.1% | 41.9 $m^3$/year | $ 0.00 | $ 918.97 | -163.7 kg CO2e | 0.688 |

The references correspond to 3 distinct configurations between **1,000 and 14,000 L**. Within this set, savings range from **19.1% to 28.0%**, CAPEX from **$ 0.00 to $ 0.00**, NPV from **$ 918.97 to $ 1,343.38**, and the climate balance from **-626.3 to -163.7 kg CO2e**. These ranges are the core of the decision; no single indicator represents all objectives.

## 1.2 How to read this report

Each row represents a physical tank configuration under the same rainfall, demand, tariff, and LCA assumptions. When two criteria point to the same volume, their roles are combined in one row. NPV and CO2e values may be negative; that is performance information, not a presentation error.

## 2. SCOPE AND BUILDING INPUT DATA

This chapter details the physical and operational premises of the building, the historical rainfall data adopted for the location, and the normative catchment parameters according to the technical standards and regulations applicable at the project site.

| Data / Parameter | Value |
|---|---|
| **Rainfall time series** | **NewYork.csv** |
| **Roof catchment area** | **150.0 m²** |
| **Runoff coefficient (C)** | **0.85** |
| **First flush discard** | **2.0 mm** |
| **Occupants / Inhabitants** | **4 people** |
| **Per-capita daily demand** | **150 L/person/day** |
| **Rainwater target share** | **28%** |
| **Climate location / station** | **New York / USA** |
| **Climate model and scenario** | **INM-CM4 · HIST** |
| **Analyzed climate period** | **01/01/1991 a 31/12/2020** |
| **Economic horizon** | **20 years** |
| **Tariff location** | **CUSTOM** |

### 2.1 Physical configuration shared by alternatives

| Component / condition | Recorded value |
|---|---|
| Upper tank in use | No |
| Upper-tank volume | 0 L |
| Nominal pump power | 1/2 CV / 0.368 kW |
| Nominal pump flow | 0 L/h |
| Pump count | 1 |
| Reported activation threshold | 10.0% |

In the comparison, lower-tank volume varies; other parameters remain constant unless scenario data explicitly state otherwise.

## 3. METHOD, BOUNDARIES AND COMPARISON CRITERIA

The study holds building assumptions constant and varies lower-tank capacity. For each capacity, Triton calculates the daily water balance, associates financial cash flows, and estimates the climate balance. Comparisons are valid only within the declared grid and boundaries.

**Hydraulic dimension**

Daily YAS model; potable savings, rainwater use, overflow, and supply reliability.

**Economic dimension**

CAPEX, tariff savings, NPV, IRR, and payback, all conditional on the time horizon and financial assumptions.

**Environmental dimension**

Simplified LCA: embodied carbon, operation, and credit for displaced potable water.

### 3.1 Criteria and effective weights

| TOPSIS criterion | Effective weight |
|---|---|
| Maximum potential | 16.7% |
| Minimum overflow | 16.7% |
| Highest NPV | 11.1% |
| Shortest discounted payback | 11.1% |
| Highest IRR | 11.1% |
| Lowest end-of-life $CO_2$ | 16.7% |
| Lowest potable-water use | 16.7% |

TOPSIS normalizes criteria and measures closeness to a constructed ideal solution. Rankings change when weights, alternatives, or data change; the first-ranked option is therefore a compromise under the stated preferences.

## 4. REPRESENTATIVE SYSTEMS AND MAIN DIFFERENCES

The configurations below are reference points for discussion, not automatic recommendations. They condense the capacities that stand out under different objectives and avoid presenting dozens of sweep rows before the reader understands the relevant differences.

| Decision role | Tank | Savings | Rainwater use | CAPEX | VPL / NPV | CO2e | TOPSIS |
|---|---|---|---|---|---|---|---|
| **Technical reference + TOPSIS compromise**<br>highest observed water-saving potential; highest weighted closeness to the ideal | **4,000 L** | 26.8% | 58.8 m³/year | $ 0.00 | $ 1,288.12 | -271.4 kg CO2e | 0.912 |
| **Economic reference**<br>highest NPV over the horizon | **14,000 L** | 28.0% | 61.3 m³/year | $ 0.00 | $ 1,343.38 | -626.3 kg CO2e | 0.840 |
| **Environmental reference**<br>best life-cycle climate balance | **1,000 L** | 19.1% | 41.9 m³/year | $ 0.00 | $ 918.97 | -163.7 kg CO2e | 0.688 |

### 4.1 Observed range across systems

> The references correspond to 3 distinct configurations between **1,000 and 14,000 L**. Within this set, savings range from **19.1% to 28.0%**, CAPEX from **$ 0.00 to $ 0.00**, NPV from **$ 918.97 to $ 1,343.38**, and the climate balance from **-626.3 to -163.7 kg CO2e**. These ranges are the core of the decision; no single indicator represents all objectives.

### 4.2 Reading rule

- Technical reference: prioritizes observed hydraulic performance; it does not imply the best financial return.
- Economic reference: prioritizes the highest NPV under current assumptions; it may reduce hydraulic or environmental performance.
- Environmental reference: prioritizes the best modeled climate balance; it depends strongly on boundaries and emission factors.
- TOPSIS compromise: combines criteria according to weights; it does not replace engineering justification.

## 5. HYDRAULIC PERFORMANCE COMPARISON (YAS MODEL)

Computational analysis of continuous daily tank simulation (YAS Algorithm — Yield After Spillage). Presents operational water supply indicators and sequential layout of hydraulic performance figures.

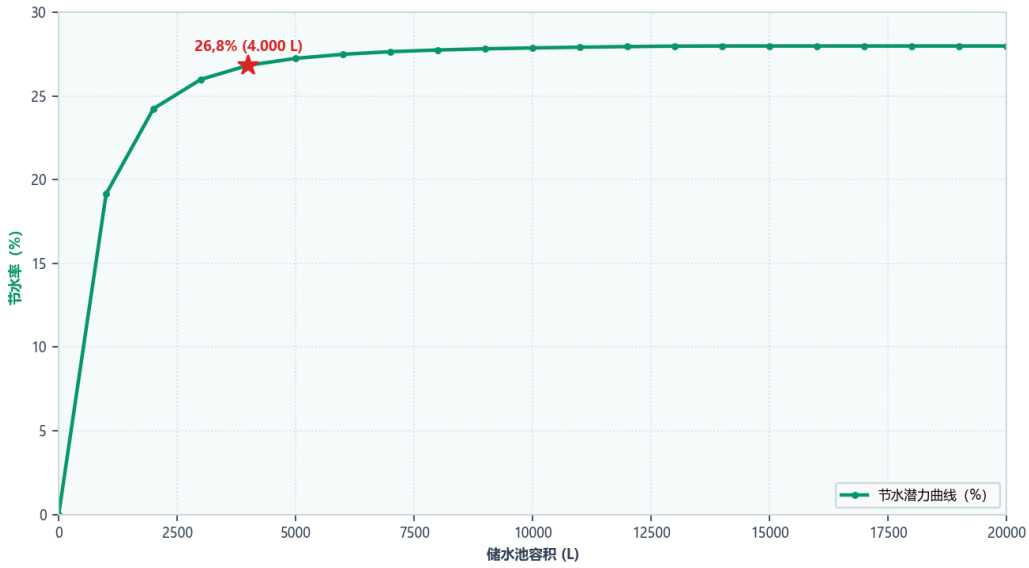
| System / role | Tank | Savings | Rainwater use | Potable water | Overflow | Full supply |
|---|---|---|---|---|---|---|
| **Technical reference + TOPSIS compromise** | 4,000 L | 26.8% | 58.8 $m^3$/year | 160.4 $m^3$/year | 53.2 $m^3$/year | 95.3% |
| **Economic reference** | 14,000 L | 28.0% | 61.3 $m^3$/year | 157.8 $m^3$/year | 50.3 $m^3$/year | 99.9% |
| **Environmental reference** | 1,000 L | 19.1% | 41.9 $m^3$/year | 177.2 $m^3$/year | 70.2 $m^3$/year | 62.2% |

### 📊 Technical Discussion on Hydrological Water Balance (YAS Model)

The continuous daily YAS (Yield After Spillage) simulation tracks storage over the climate series. A larger tank tends to increase rainwater use and reduce overflow, but marginal gains decline as rainfall, catchment area, and demand become limiting factors. Read the table horizontally: it shows the hydraulic gain or loss when moving among references without assuming that the largest tank is the best overall choice.

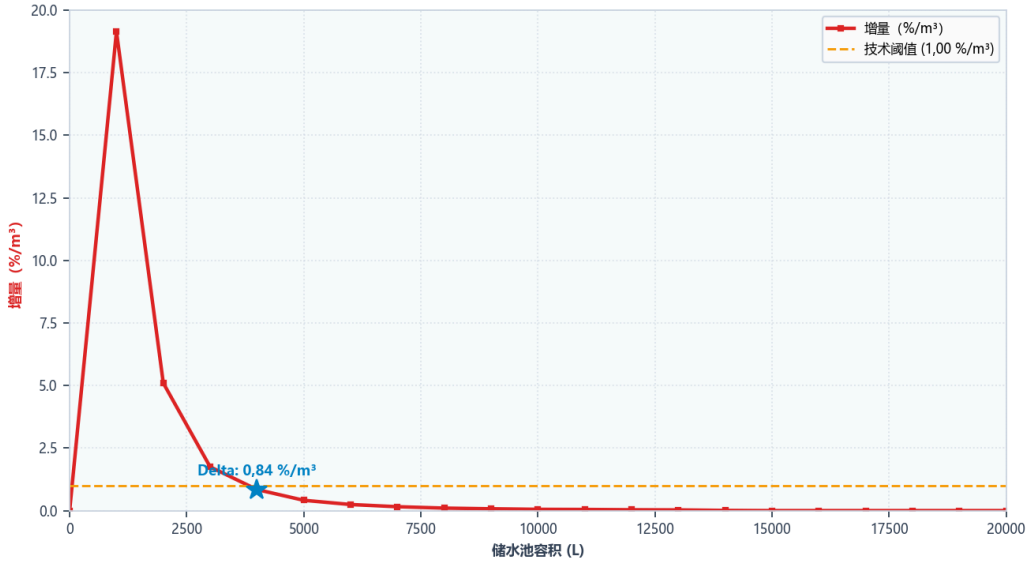
## Figures generated for this chapter

### Figure 1 — Hydraulic performance: water-saving potential



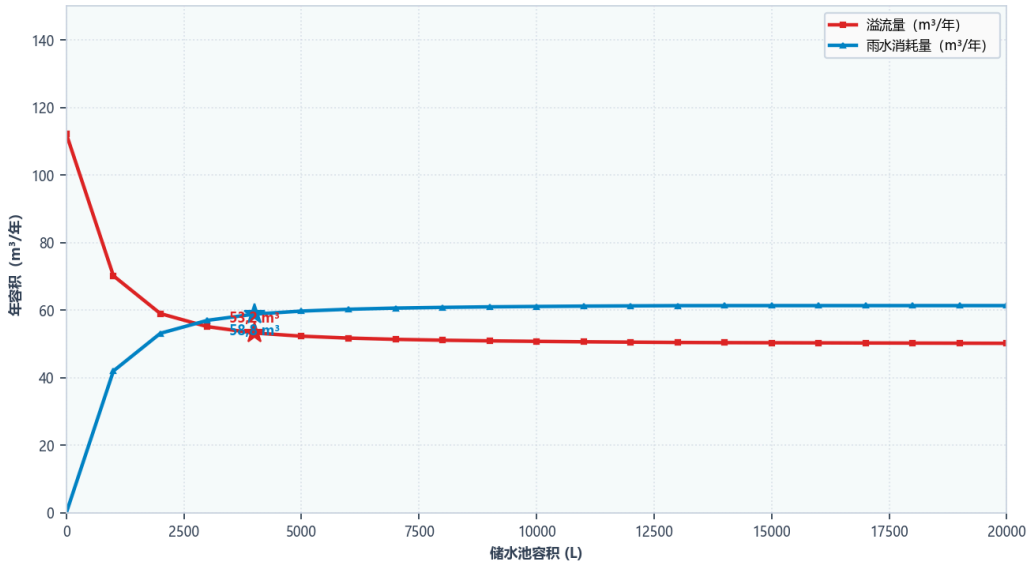

**Reading:**
Shows the gain in potable-water substitution as storage capacity increases.
Source: results calculated by Triton from this study's inputs.

### Figure 2 — Hydraulic performance: marginal gain



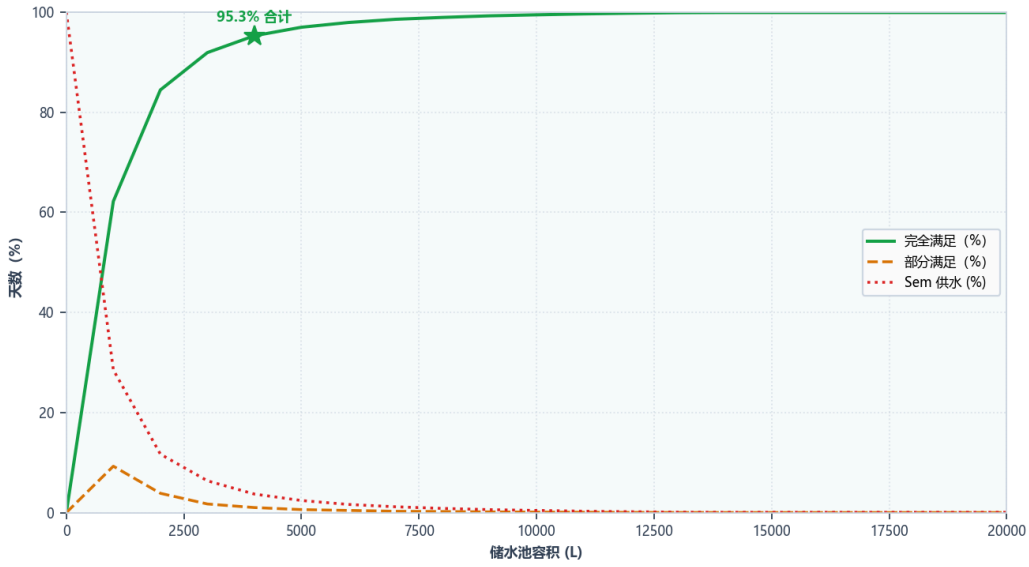

**Reading:**
Highlights diminishing returns and where additional tank capacity yields little extra hydraulic benefit.
Source: results calculated by Triton from this study's inputs.

## Figure 3 — Hydraulic performance: rainwater use and overflow




**Reading:**
Compares effectively used rainwater with overflow at each evaluated capacity.
Source: results calculated by Triton from this study's inputs.

## Figure 4 — Hydraulic performance: annual supply reliability




**Reading:**
Presents the frequency of full, partial, and zero supply over the simulated series.
Source: results calculated by Triton from this study's inputs.

## 6. MULTI-CRITERIA DECISION, RANKING AND TRADE-OFFS

Multi-criteria evaluation of swept alternatives. The ranking aggregates preferences; difference tables preserve physical and economic values that the normalized score alone does not communicate.

### 6.1 Objective-based references and weighted compromise

| Decision role | Tank | Savings | Rainwater use | CAPEX | VPL / NPV | CO2e | TOPSIS |
|---|---|---|---|---|---|---|---|
| **Technical reference + TOPSIS compromise**<br>highest observed water-saving potential; highest weighted closeness to the ideal | **4,000 L** | 26.8% | 58.8 m³/year | $ 0.00 | $ 1,288.12 | -271.4 kg CO2e | 0.912 |
| **Economic reference**<br>highest NPV over the horizon | **14,000 L** | 28.0% | 61.3 m³/year | $ 0.00 | $ 1,343.38 | -626.3 kg CO2e | 0.840 |
| **Environmental reference**<br>best life-cycle climate balance | **1,000 L** | 19.1% | 41.9 m³/year | $ 0.00 | $ 918.97 | -163.7 kg CO2e | 0.688 |

⚖️ **Multi-Criteria Decision Discussion (TOPSIS & Trade-offs)**

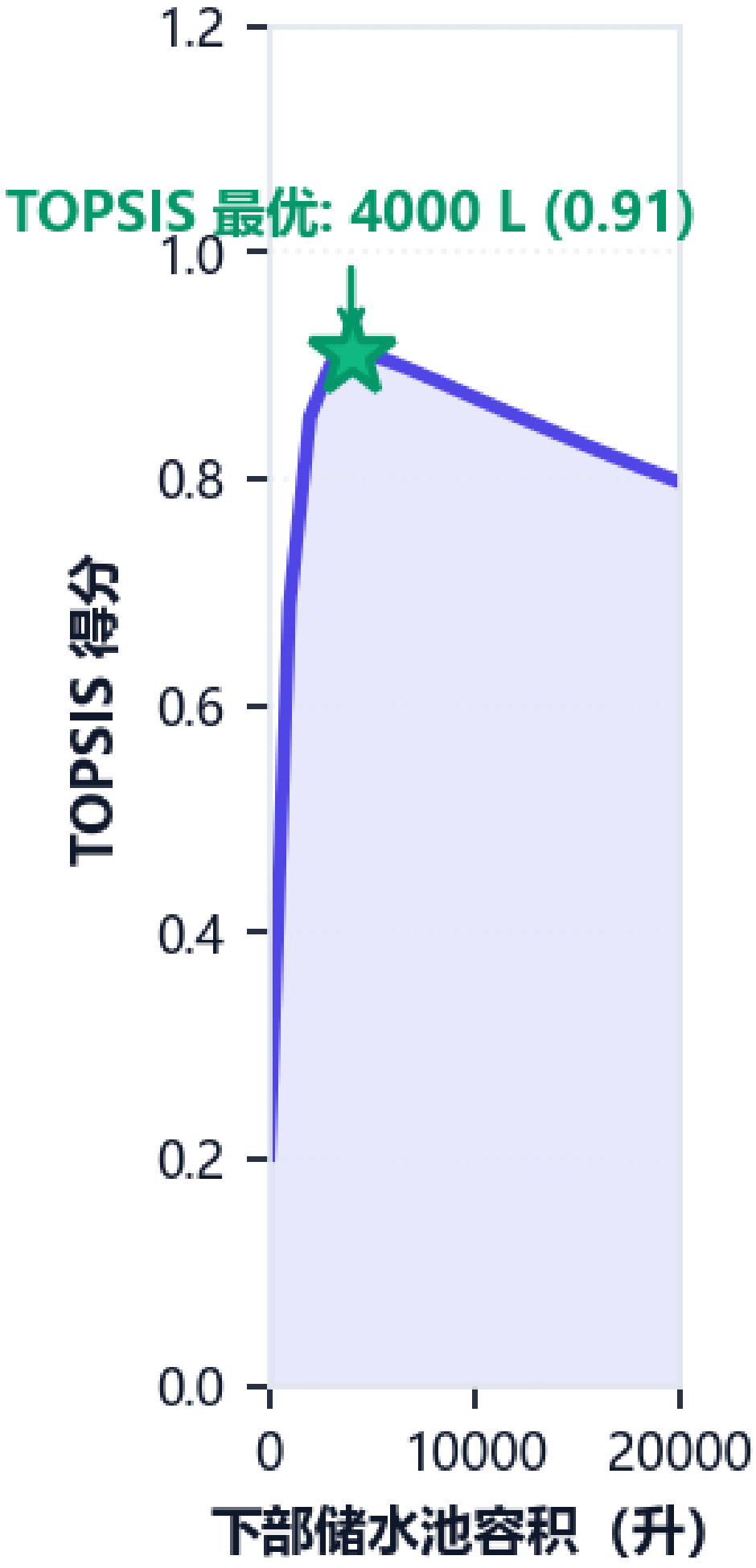
The **technical reference** represents the hydraulic highlight of the set; the **economic reference**, the highest NPV; the **environmental reference**, the best modeled climate balance; and the **topsis compromise**, highest weighted closeness to the ideal. The references correspond to 3 distinct configurations between **1,000 and 14,000 L**. Within this set, savings range from **19.1% to 28.0%**, CAPEX from **$ 0.00 to $ 0.00**, NPV from **$ 918.97 to $ 1,343.38**, and the climate balance from **-626.3 to -163.7 kg CO2e**. These ranges are the core of the decision; no single indicator represents all objectives.

### 6.2 Decision matrix and ranking of all alternatives

| Tank Volume | Savings (%) | Rainwater Use | CAPEX | NPV (20 yrs) | CO2e Saved | TOPSIS Score | Rank |
|---|---|---|---|---|---|---|---|
| **0 L** | 0.0% | 0.0 m³/yr | $ 0.00 | $ 0.00 | 0.0 kg | 0.20 | **#21** |
| **1,000 L** | 19.1% | 41.9 m³/yr | $ 0.00 | $ 918.97 | -163.7 kg | 0.69 | **#20** |
| **2,000 L** | 24.2% | 53.1 m³/yr | $ 0.00 | $ 1,163.69 | -200.1 kg | 0.85 | **#11** |
| **3,000 L** | 26.0% | 57.0 m³/yr | $ 0.00 | $ 1,247.69 | -235.8 kg | 0.90 | **#4** |
| **4,000 L** | **26.8%** | **58.8 m³/yr** | **$ 0.00** | **$ 1,288.12** | **-271.4 kg** | **0.91** | **#1** |
| **5,000 L** | 27.2% | 59.7 m³/yr | $ 0.00 | $ 1,307.93 | -307.0 kg | 0.91 | **#2** |
| **6,000 L** | 27.5% | 60.2 m³/yr | $ 0.00 | $ 1,319.63 | -342.5 kg | 0.90 | **#3** |
| **7,000 L** | 27.6% | 60.6 m³/yr | $ 0.00 | $ 1,327.11 | -378.0 kg | 0.90 | **#5** |
| **8,000 L** | 27.7% | 60.8 m³/yr | $ 0.00 | $ 1,331.93 | -413.4 kg | 0.89 | **#6** |
| **9,000 L** | 27.8% | 61.0 m³/yr | $ 0.00 | $ 1,335.41 | -448.9 kg | 0.88 | **#7** |
| **10,000 L** | 27.9% | 61.1 m³/yr | $ 0.00 | $ 1,337.83 | -484.4 kg | 0.87 | **#8** |
| **11,000 L** | 27.9% | 61.2 m³/yr | $ 0.00 | $ 1,340.02 | -519.9 kg | 0.86 | **#9** |
| **12,000 L** | 27.9% | 61.2 m³/yr | $ 0.00 | $ 1,341.66 | -555.3 kg | 0.86 | **#10** |
| **13,000 L** | 28.0% | 61.3 m³/yr | $ 0.00 | $ 1,343.00 | -590.8 kg | 0.85 | **#12** |
| **14,000 L** | 28.0% | 61.3 m³/yr | $ 0.00 | $ 1,343.38 | -626.3 kg | 0.84 | **#13** |
| **15,000 L** | 28.0% | 61.3 m³/yr | $ 0.00 | $ 1,343.38 | -661.7 kg | 0.83 | **#14** |
| **16,000 L** | 28.0% | 61.3 m³/yr | $ 0.00 | $ 1,343.38 | -697.2 kg | 0.83 | **#15** |
| **17,000 L** | 28.0% | 61.3 m³/yr | $ 0.00 | $ 1,343.38 | -732.7 kg | 0.82 | **#16** |
| **18,000 L** | 28.0% | 61.3 m³/yr | $ 0.00 | $ 1,343.38 | -768.1 kg | 0.81 | **#17** |
| **19,000 L** | 28.0% | 61.3 m³/yr | $ 0.00 | $ 1,343.38 | -803.6 kg | 0.80 | **#18** |
| **20,000 L** | 28.0% | 61.3 m³/yr | $ 0.00 | $ 1,343.38 | -839.1 kg | 0.80 | **#19** |

## Figures generated for this chapter

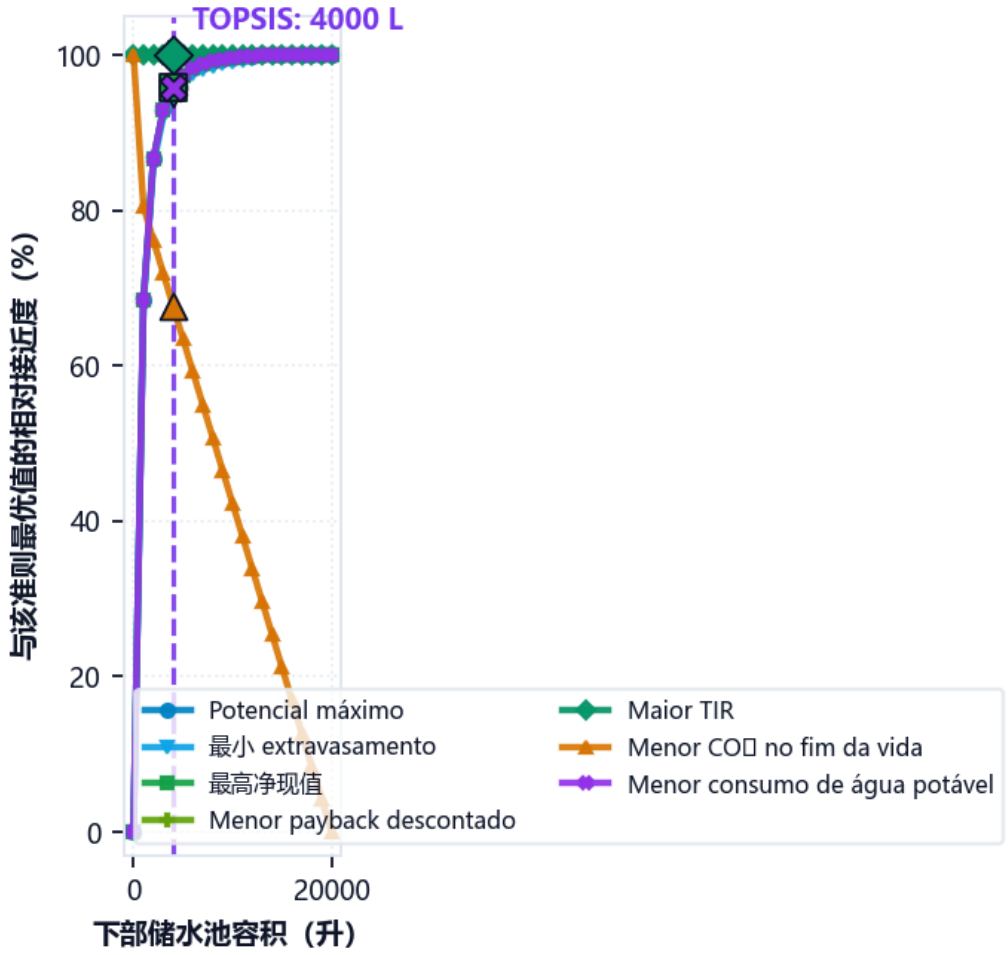
**Figure 5 — Multi-criteria decision: TOPSIS performance**




**Reading:**
Shows each alternative's compromise score under the stated weights; small differences indicate proximity, not absolute

superiority.
Source: results calculated by Triton from this study's inputs.

**Figure 6 — Multi-criteria decision: proximity by dimension**




**Reading:**
Separates proximity to technical, economic, and environmental references so trade-offs remain visible.
Source: results calculated by Triton from this study's inputs.

## 7. ECONOMIC AND FINANCIAL COMPARISON

Economic-financial feasibility study based on Discounted Cash Flow (DCF) over a 20-year analytical horizon, incorporating tariff escalation and discount rates.

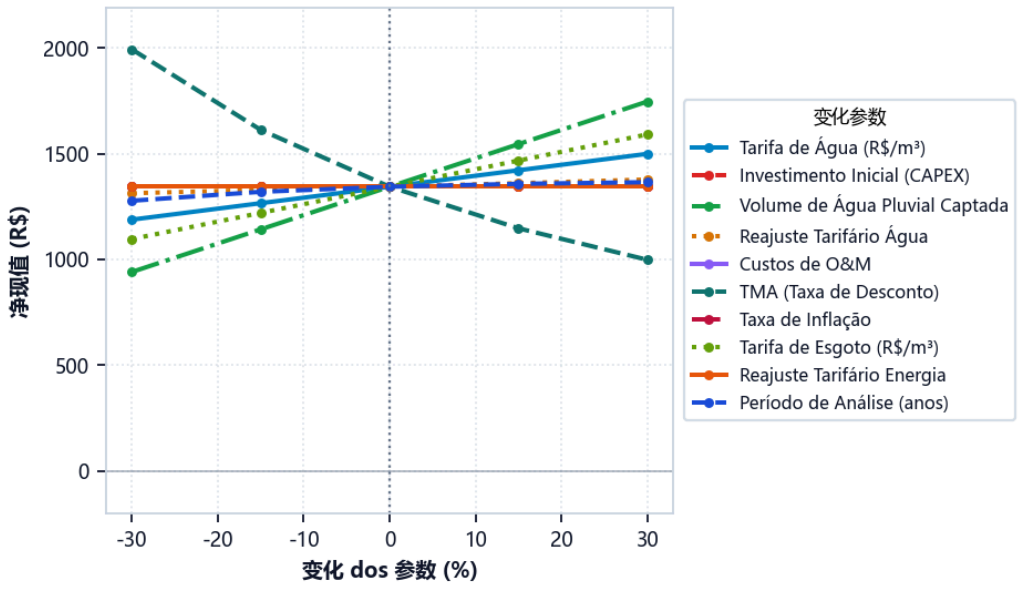
| System / role | Tank | CAPEX | VPL / NPV | Gap to highest NPV | Discounted payback | TIR / IRR |
|---|---|---|---|---|---|---|
| **Technical reference + TOPSIS compromise** | 4,000 L | $ 0.00 | $ 1,288.12 | $ -55.25 | -- | 0.0% |
| **Economic reference** | 14,000 L | $ 0.00 | $ 1,343.38 | $ 0.00 | -- | 0.0% |
| **Environmental reference** | 1,000 L | $ 0.00 | $ 918.97 | $ -424.41 | -- | 0.0% |

### 💰 Technical Discussion on Financial Feasibility & Cash Flow

The analysis uses Discounted Cash Flow over the reported horizon. The gap-to-highest-NPV column quantifies the financial opportunity cost of prioritizing another objective. Positive NPV indicates a return above the discount rate under the stated assumptions; negative NPV means the modeled opportunity cost is not recovered. IRR and payback must be compared with the hurdle rate, horizon, and parameter risk without automatic investment-approval language. Official municipal tariffs audited and validated on 27/08/2026 (confirming recent regulatory updates is recommended).

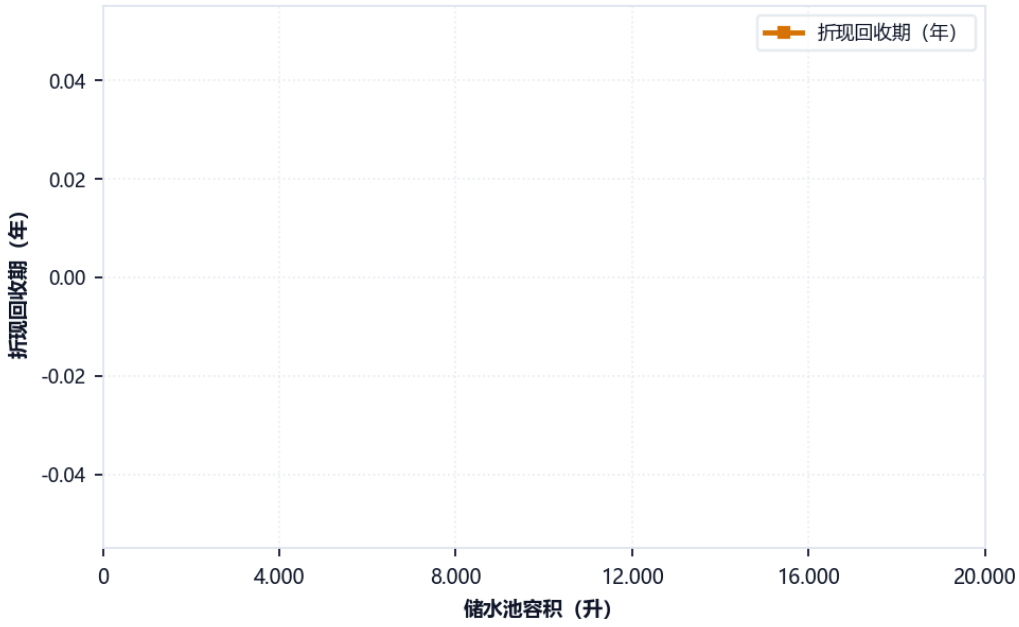
## Figures generated for this chapter

### Figure 7 — Economic assessment: NPV sensitivity




**Reading:**
Indicates which financial assumptions most affect NPV and therefore deserve priority validation.
Source: results calculated by Triton from this study's inputs.

### Figure 8 — Economic assessment: NPV by capacity




**Reading:**
Compares financial returns across capacities without confusing greater water savings with higher NPV.
Source: results calculated by Triton from this study's inputs.

## 8. ENVIRONMENTAL COMPARISON BY SIMPLIFIED LCA

Simplified Life Cycle Assessment (LCA) calculating embodied carbon emissions vs. operational avoided emissions from municipal water supply.

### 8.1 Climate-balance comparison across systems

| System / role | Tank | Life-cycle balance | Gap to best balance | Interpretation |
|---|---|---|---|---|
| **Technical reference + TOPSIS compromise** | 4,000 L | -271.4 kg CO2e | -107.7 kg CO2e | modeled net impact |
| **Economic reference** | 14,000 L | -626.3 kg CO2e | -462.6 kg CO2e | modeled net impact |
| **Environmental reference** | 1,000 L | -163.7 kg CO2e | 0.0 kg CO2e | modeled net impact |

### 8.2 Detailed inventory for the report reference alternative

| Data / Parameter | Value |
|---|---|
| **Initial embodied carbon (Embodied CO2e)** | **+621.4 kg CO2e** |
| **Annual pump operational emissions** | **+0.0 kg CO2e/yr** |
| **Calculated chlorination** | **0,142 kg Cl2/yr (1.50 mg/L)** |
| **Chlorination emissions** | **+0.24 kg CO2e/yr (0,004 kg CO2e/m³)** |
| **Annual avoided public grid emissions** | **-0.0 kg CO2e/yr** |
| **Annual net avoided carbon balance** | **+0.2 kg CO2e/yr** |
| **Cumulative 20-year net CO2e balance** | **+626.3 kg CO2e** |
| **Environmental payback period** | **999.0 years** |

**Technical Discussion on Life Cycle Assessment (LCA)**

The simplified LCA is a comparative screening assessment. Under Triton's official convention, atmospheric emissions are positive (+) in red (environmental impact), whereas avoided emissions or removed carbon are negative (-) in green (ecological benefit). For the detailed alternative, initial embodied carbon is **+621.4 kg CO2e** and calculated payback is **999.0 years**. These figures do not certify sustainability; they depend on quantities, factors, boundaries, service life, and the credit assigned to displaced water.

**20-Year Lifetime Cumulative Net Carbon Trajectory**

YEAR 0 — INITIAL FOOTPRINT (CRADLE-TO-GATE)

**-621.4 kg CO2e**

Embodied manufacturing and transport emissions

YEAR 20 — CUMULATIVE NET IMPACT

**-626.3 kg CO2e**

Embodied and operational impacts exceed the modeled credit

## 9. SENSITIVITY, UNCERTAINTY AND LIMITATIONS

The numerical differences in this report are conditional, not deterministic forecasts. Selection robustness depends on whether the ranking and the signs of NPV/CO2e persist under plausible input changes.

- **Climate and historical series:** record length, gaps, representativeness, and climate change may alter reliability and overflow.
- **Demand and behavior:** occupancy, non-potable uses, and seasonality should be tested under low, central, and high scenarios.
- **Economics:** CAPEX, maintenance, tariffs, escalation, and discount rate may reverse NPV, IRR, and payback.
- **LCA:** material mass, electricity source, pump efficiency, displaced treatment, and service life control the climate balance.
- **Multi-criteria decision:** weights express stated preferences; rerun TOPSIS with alternative weights and record changes in the first-ranked option.

> Suggested minimum robustness criterion: the selected alternative should remain technically acceptable under adverse scenarios, maintain economic indicators compatible with the decision maker's policy, and not depend on a single environmental factor without a traceable source.

## 10. CONCLUSIONS AND SELECTION PROTOCOL

The references correspond to 3 distinct configurations between **1,000 and 14,000 L**. Within this set, savings range from **19.1% to 28.0%**, CAPEX from **$ 0.00 to $ 0.00**, NPV from **$ 918.97 to $ 1,343.38**, and the climate balance from **-626.3 to -163.7 kg CO2e**. These ranges are the core of the decision; no single indicator represents all objectives.

### 10.1 Recommended protocol for documenting the decision

| Step | Question | Expected record |
|---|---|---|
| **1** | Which constraints are non-negotiable? | Space, commercial size, budget, minimum service, and regulatory requirements. |
| **2** | Which objective has priority and why? | Technical, economic, environmental, or compromise rationale; approved TOPSIS weights. |
| **3** | What is the cost of moving to another reference? | Differences in tank size, water savings, CAPEX, NPV, and CO2e from the comparison tables. |
| **4** | Is the choice robust to uncertainty? | Scenarios tested, critical parameters, ranking changes, and safety margins. |
| **5** | Which configuration will be adopted? | Final capacity, components, assumptions, responsible party, date, and signed rationale. |

Triton supports this decision but does not replace detailed design, regulatory verification, risk analysis, or professional responsibility.

## 11. TRACEABILITY, REFERENCES AND PROFESSIONAL RESPONSIBILITY

- **ISO 14040/14044** — Environmental management — Life cycle assessment — Principles and framework.
- **Emission factors of the grid and of the water supply** — Values entered by the user for the project site; record the source, the year and the system boundary.
- **Materials and equipment factors** — Screening parameters; document supplier EPD/inventory, version, boundary and dataset.
- **Hwang & Yoon (1981)** — *Multiple Attribute Decision Making*; fundamento do TOPSIS.

> ℹ️ **User Responsibility for Equation Verification**
> The report records implemented equations and assumptions to support audit. The study team must confirm local data, sources, boundaries, standards and results before making a decision; internal tests are not external validation.

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_____
**Responsible Researcher / Engineer**
Professional registration / Licence number